Code No: 136CX **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August/September - 2024
MOBILE COMPUTING
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Used paper Aug-2024

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	Explain the types of mobility.	[2]
b)	Explain the reference model of Bearer services of GSM with neat sketch.	[3]
c)	What is tunneling?	[2]
d)	What is the use of DHCP protocol?	[3]
e)	What are the advantages of Mobile TCP?	[2]
f)	Explain the term hoarding.	[3]
g)	What is meant by Stateless Synchronous?	[2]
h)	Explain the reasons why data synchronization is required in mobile computing.	[3]
i)	What is the importance of Windows CE operating system?	[2]
j)	Explain the baseband layer of Bluetooth.	[3]

PART – B

(50 Marks)

2.a)	Explain the characteristics of mobile communication device.	
b)	Explain the TDMA frame structure of GSM.	[5+5]
	OR	
3.a)	Explain the differences between bearer and the services of GSM.	
b)	Explain the protocol architecture of GSM with signaling protocols.	[5+5]
4.a)	Write about the power management in IEEE 802.11 infrastructure networks.	
b)	Write about the data delivery using mobile IP.	[5+5]
	OR	
5.a)	Explain the basic DFWMAC – DCF using RTS/CTS.	
b)	Write about the generic routing encapsulation.	[5+5]
6.a)	Explain how indirect TCP hides the problems of wireless links from fixed host.	
b)	Give the architecture for Query Processing in mobile networks.	[5+5]
	OR	
7.a)	Write about the transaction-oriented TCP.	
b)	Explain about the architecture of three-tier client - server computing in mobile networks.	[5+5]

8. What is meant by broad cast disk model? [10]

OR

9.a) Explain about the Pull based data delivery mechanism with neat sketch.

b) Write about the distributed index-based method. [5+5]

10.a) Explain the steps involved in basic transaction of WTP class 2 with user Acknowledgement.

b) Explain the challenges of service discovery in mobile environment. [5+5]

OR

11. Explain how to find the route from source to destination by using DSDV protocol and also illustrate with an example. [10]

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